

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\mathcal{J}_D(\delta) = \inf_{\substack{\lambda \in \mathbb{R}_+^L, \psi_\ell > -\infty \\ (\psi_\ell)_{\ell \in [L]} \in \prod_{\ell \in [L]} L^1(\mu_\ell)}} \left\{ \langle \lambda, \delta \rangle + \sum_{\ell=1}^L \int \psi_\ell d\mu_\ell : \sum_{\ell=1}^L \psi_\ell(s_\ell) \geq g_{\lambda, [L]}(s), \forall s \right\}.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$C_n(x) = \begin{cases} K - vx + G_n(x + B) & s_{k-1} < x \leq \hat{S}_k - B \quad k = 1, \dots, w-1 \\ \textit{beginalign*} - 4pt] K - vx + G_n(\hat{S}_k) & \hat{S}_k - B < x \leq s_k \quad k = 1, \dots, w-1 \\ \textit{beginalign*} - 4pt] K - vx + G_n(x + B) & s_{w-1} < x \leq S_m - B \\ \textit{beginalign*} - 4pt] K - vx + G_n(S_m) & S_m - B < x \leq s_m \\ \textit{beginalign*} - 4pt] - vx + G_n(x) & x > s_m. \end{cases}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \pm \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \pm \frac{q_i}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & |\phi_n(\alpha_a(E_\psi)) - \chi(E_\psi)| \\ &= |\phi_n(\alpha_a(E_\psi)) - \chi(E_{f_n} E_\psi)| \\ &\leq \int_{b \in U_n} f_n(b) \left(\int |\Delta(b^{-1}a)^m \psi(g_1 b^{-1}a, \dots, g_m b^{-1}a) - \psi(g_1, g_2, \dots, g_m)| dg_1 \cdots dg_m \right) db \\ &\leq \epsilon \int_{b \in U_n} f_n(b) \\ &\leq \epsilon. \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
V^{(1)} = & \frac{1}{4!}f\varphi^4 + \frac{1}{48}(\beta_f - 4f\gamma_\varphi)\varphi^4 \left(\log \frac{\varphi^2}{\mu^2} - \frac{25}{6} \right) \\
& - \frac{1}{2}\xi R\varphi^2 - \frac{1}{4}(\beta_\xi - 2\xi\gamma_\varphi)R\varphi^2 \left(\log \frac{\varphi^2}{\mu^2} - 3 \right),
\end{aligned}$$